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Bihar Engineering University, Patna

B.Tech 2nd Semester Examination, 2025 (Old Course)

Time: 03 Hours

Code: 100205 ***Subject: Workshop Manufacturing Practices*** ***Full Marks: 70***

Instructions:-

(i) The marks are indicated in the right-hand margin.
(ii) There are NINE questions in this paper.
(iii) Attempt FIVE questions in all.
(iv) Question No. 1 is compulsory.

Q.1 Choose the correct option/answer the following (Any seven question only): **[14]**

(a) Which of the following processes is primarily used for producing complex shapes with internal cavities?

(i) Machining (iii) Forming
(ii) Casting (iv) Joining

(b) In advanced manufacturing methods, which technology uses a layer-by-layer approach to build parts?

(i) CNC Machining (iii) Welding
(ii) Additive Manufacturing (iv) Forging

(c) A hacksaw blade is specified by its

(i) Length (iii) Material
(ii) Width (iv) Number of teeth

(d) Which tool is commonly used for tightening or loosening nuts and bolts during fitting operations?

(i) Hacksaw (iii) File
(ii) Spanner (iv) Hammer

(e) Which of the following is a power tool used for drilling holes in metal or wood?

(i) Bench vice (iii) Hand file
(ii) Electric drill (iv) Chisel

(f) In fitting operations, which instrument is used to check the flatness and alignment of surfaces?

(i) Vernier caliper (iii) Micrometer
(ii) Surface plate (iv) Dial gauge

(g) Which of the following is the primary material used in carpentry?

(i) Metal (iii) Plastic
(ii) Wood (iv) Ceramic

(h) In carpentry, which joint is most commonly used for connecting two wooden pieces at right angles?

(i) Lap joint (iii) Mortise and tenon joint
(ii) Butt joint (iv) Dovetail joint

(i) Which of the following processes is used to shape molten plastic into desired forms?

(i) Injection moulding (iii) Casting
(ii) Forging (iv) Extrusion of metals

(j) Which of the following joining methods is suitable for joining thin electrical wires?

(i) Arc welding (iii) Brazing
(ii) Gas welding (iv) Soldering

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Q.2 Solve both questions : **[14]**

(a) What is casting? Explain any one type of casting process with a neat sketch.

(b) Differentiate between hot working and cold working processes. Discuss their advantages and limitations.

Q.3 Solve both questions : **[14]**

(a) Describe the forging process. How does it differ from extrusion?

(b) Discuss the factors affecting metal forming operations such as material properties, temperature, and strain rate.

Q.4 Solve both questions : **[14]**

(a) Explain the rolling process in detail. What are its applications in the metal forming industry?

(b) What is additive manufacturing? Describe its applications in modern industries.

Q.5 Solve both questions : **[14]**

(a) Explain the working principle of CNC machining. Discuss its advantages over conventional machining.

(b) Compare CNC machining and additive manufacturing in terms of material utilization, complexity of shapes, and cost.

Q.6 Solve both questions : **[14]**

(a) What is carpentry? Explain its importance in construction and furniture making.

(b) Describe any three common carpentry joints with neat sketches.

Q.7 Solve both questions : **[14]**

(a) What is plastic moulding? Explain any one type of plastic moulding process with a neat sketch.

(b) Describe the procedure of glass cutting and list safety precautions to be followed.

Q.8 Solve both questions : **[14]**

(a) Compare arc welding and gas welding in terms of working principle, applications, and advantages.

(b) What is brazing? How does it differ from soldering? Give examples of applications for both.

Q.9 Solve both questions : **[14]**

(a) Write short notes on: Flux in welding.

(b) Write short notes on: Heat-affected zone in welding.

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